KAIZEN MATHS

Lesson starter

Partial Fractions: Decomposition

Quadratic Factors

Name _____

S1

Express in partial fractions:

$\frac{14x+15}{(x-3)(x^2+10)}$

S2

Express in partial fractions:

$\frac{-x^2+4x-14}{x(x^2+7)}$

S3

Express in partial fractions:

$\frac{3x^2-x+28}{(x-4)(x^2+2)}$

S4

Express in partial fractions:

$\frac{3x^2+2x+11}{(x+1)(x^2+2)}$

Show your working

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ALG-36C3-L4 · S

Show your working

KAIZEN MATHS

Exit ticket

Partial Fractions: Decomposition

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E1

Express in partial fractions:

$\frac{6x^2-11x+35}{(x-3)(x^2+5)}$

E2

Express in partial fractions:

$\frac{-5x^2+2x-18}{(x-3)(x^2+10)}$

E3

Express in partial fractions:

$\frac{2x^2-8x+17}{(x-3)(x^2+2)}$

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Teacher answers & working

Partial Fractions: Decomposition · Quadratic Factors

ALG-36C3-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Express in partial fractions:

$\frac{14x+15}{(x-3)(x^2+10)}$

Answer:

$\frac{3}{(x-3)} + \frac{-3x+5}{x^2+10}$

Working:

Use a constant over the linear factor and a linear numerator over the quadratic factor.

$\frac{14x+15}{(x-3)(x^2+10)} = \frac{A}{(x-3)} + \frac{Bx+C}{x^2+10}$

Multiply through by (x - 3)(x^2 + 10).

$14x + 15 = A(x^2 + 10) + (Bx + C)(x - 3)$

Expand the right hand side.

$14x + 15 = (A + B)x^2 + (C - 3B)x + (10A - 3C)$

Compare coefficients: A + B = 0, C - 3B = 14, and 10A - 3C = 15.

Solving these gives A = 3, B = -3, and C = 5.

Therefore $\frac{14x+15}{(x-3)(x^2+10)} = \frac{3}{(x-3)} + \frac{-3x+5}{x^2+10}$.

S2

Express in partial fractions:

$\frac{-x^2+4x-14}{x(x^2+7)}$

Answer:

$\frac{-2}{x} + \frac{x+4}{x^2+7}$

Working:

Use a constant over the linear factor and a linear numerator over the quadratic factor.

$\frac{-x^2+4x-14}{x(x^2+7)} = \frac{A}{x} + \frac{Bx+C}{x^2+7}$

Multiply through by x(x^2 + 7).

$-x^2 + 4x - 14 = A(x^2 + 7) + (Bx + C)x$

Expand the right hand side.

$-x^2 + 4x - 14 = (A + B)x^2 + (C - 0B)x + (7A - 0C)$

Compare coefficients: A + B = -1, C - 0B = 4, and 7A - 0C = -14.

Solving these gives A = -2, B = 1, and C = 4.

Therefore $\frac{-x^2+4x-14}{x(x^2+7)} = \frac{-2}{x} + \frac{x+4}{x^2+7}$.

Teacher answers & working

Partial Fractions: Decomposition · Quadratic Factors

ALG-36C3-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

Express in partial fractions:

$\frac{3x^2-x+28}{(x-4)(x^2+2)}$

Answer:

$\frac{4}{(x-4)} + \frac{-x-5}{x^2+2}$

Working:

Use a constant over the linear factor and a linear numerator over the quadratic factor.

$\frac{3x^2-x+28}{(x-4)(x^2+2)} = \frac{A}{(x-4)} + \frac{Bx+C}{x^2+2}$

Multiply through by $(x - 4)(x^2 + 2)$.

$3x^2 - x + 28 = A(x^2 + 2) + (Bx + C)(x - 4)$

Expand the right hand side.

$3x^2 - x + 28 = (A + B)x^2 + (C - 4B)x + (2A - 4C)$

Compare coefficients: A + B = 3, C - 4B = -1, and 2A - 4C = 28.

Solving these gives A = 4, B = -1, and C = -5.

Therefore $\frac{3x^2-x+28}{(x-4)(x^2+2)} = \frac{4}{(x-4)} + \frac{-x-5}{x^2+2}$.

S4

Express in partial fractions:

$\frac{3x^2+2x+11}{(x+1)(x^2+2)}$

Answer:

$\frac{4}{(x+1)} + \frac{-x+3}{x^2+2}$

Working:

Use a constant over the linear factor and a linear numerator over the quadratic factor.

$\frac{3x^2+2x+11}{(x+1)(x^2+2)} = \frac{A}{(x+1)} + \frac{Bx+C}{x^2+2}$

Multiply through by $(x + 1)(x^2 + 2)$.

$3x^2 + 2x + 11 = A(x^2 + 2) + (Bx + C)(x + 1)$

Expand the right hand side.

$3x^2 + 2x + 11 = (A + B)x^2 + (C + B)x + (2A + C)$

Compare coefficients: A + B = 3, C + B = 2, and 2A + C = 11.

Solving these gives A = 4, B = -1, and C = 3.

Therefore $\frac{3x^2+2x+11}{(x+1)(x^2+2)} = \frac{4}{(x+1)} + \frac{-x+3}{x^2+2}$.

Teacher answers & working

Partial Fractions: Decomposition · Quadratic Factors

ALG-36C3-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E1

Express in partial fractions:
$\frac{6x^2-11x+35}{(x-3)(x^2+5)}$

Answer:

$\frac{4}{(x-3)} + \frac{2x-5}{x^2+5}$

Working:

Use a constant over the linear factor and a linear numerator over the quadratic factor.

$\frac{6x^2-11x+35}{(x-3)(x^2+5)} = \frac{A}{(x-3)} + \frac{Bx+C}{x^2+5}$

Multiply through by $(x - 3)(x^2 + 5)$.

$6x^2 - 11x + 35 = A(x^2 + 5) + (Bx + C)(x - 3)$

Expand the right hand side.

$6x^2 - 11x + 35 = (A + B)x^2 + (C - 3B)x + (5A - 3C)$

Compare coefficients: $A + B = 6$, $C - 3B = -11$, and $5A - 3C = 35$.

Solving these gives $A = 4$, $B = 2$, and $C = -5$.

Therefore $\frac{6x^2-11x+35}{(x-3)(x^2+5)} = \frac{4}{(x-3)} + \frac{2x-5}{x^2+5}$.

E2

Express in partial fractions:
$\frac{-5x^2+2x-18}{(x-3)(x^2+10)}$

Answer:

$\frac{-3}{(x-3)} + \frac{-2x-4}{x^2+10}$

Working:

Use a constant over the linear factor and a linear numerator over the quadratic factor.

$\frac{-5x^2+2x-18}{(x-3)(x^2+10)} = \frac{A}{(x-3)} + \frac{Bx+C}{x^2+10}$

Multiply through by $(x - 3)(x^2 + 10)$.

$-5x^2 + 2x - 18 = A(x^2 + 10) + (Bx + C)(x - 3)$

Expand the right hand side.

$-5x^2 + 2x - 18 = (A + B)x^2 + (C - 3B)x + (10A - 3C)$

Compare coefficients: $A + B = -5$, $C - 3B = 2$, and $10A - 3C = -18$.

Solving these gives $A = -3$, $B = -2$, and $C = -4$.

Therefore $\frac{-5x^2+2x-18}{(x-3)(x^2+10)} = \frac{-3}{(x-3)} + \frac{-2x-4}{x^2+10}$.

Teacher answers & working

Partial Fractions: Decomposition · Quadratic Factors

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E3

Express in partial fractions:

$\frac{2x^2-8x+17}{(x-3)(x^2+2)}$

Answer:

$\frac{1}{(x-3)} + \frac{x-5}{x^2+2}$

Working:

Use a constant over the linear factor and a linear numerator over the quadratic factor.

$\frac{2x^2-8x+17}{(x-3)(x^2+2)} = \frac{A}{(x-3)} + \frac{Bx+C}{x^2+2}$

Multiply through by $(x - 3)(x^2 + 2)$.

$2x^2 - 8x + 17 = A(x^2 + 2) + (Bx + C)(x - 3)$

Expand the right hand side.

$2x^2 - 8x + 17 = (A + B)x^2 + (C - 3B)x + (2A - 3C)$

Compare coefficients: $A + B = 2$, $C - 3B = -8$, and $2A - 3C = 17$.

Solving these gives $A = 1$, $B = 1$, and $C = -5$.

Therefore $\frac{2x^2-8x+17}{(x-3)(x^2+2)} = \frac{1}{(x-3)} + \frac{x-5}{x^2+2}$.